ACTU 301
Life Contingencies 1

Life Tables

Lecture 3

Life Table Functions

A life table is a tabular representation of mortality evolution for a group of lives.

Suppose that we track a cohort of lives. At the end of each year, we keep track of the number of survivors.

Initially the number of lives is denoted by l_0.This starting value is called the radix of the life table.

For x = $1, 2, ...,$ the function l_x stands for the expected number of persons who can survive to age x (or the number of individuals alive at age).

Given an assumed value of l_0 , we can express any survival function $S_0(x)$ in a tabular form by using the relation

$$l_x = l_0\, S_0(x)$$

In other words, given the life table function l_x, we can easily obtain values of $S_0(x)$ for integral values of x using the relation

$$S_0(x) = \frac{l_x}{l_0}$$

The life table may not always begin with newborn lives and instead starts at some integer age x, if this is the case we need an expression for the survival function $S_x(t)$.

We know that l_x is the number of individuals alive at age x, so l_{x+t} is the number of individuals alive at age $x+t$.

$${}_tp_x = S_x(t) = \frac{S_0(x+t)}{S_0(x)} = \frac{l_{x+t}/l_0}{l_x/l_0} = \frac{l_{x+t}}{l_x}$$

$$S_x(t) = {}_tp_x = \frac{l_{x+t}}{l_x}$$

which means that we can calculate ${}_tp_x$ for all integral values of t and x from the life table function l_x

The difference $l_x – l_{x+t}$ is the expected number of deaths over the age interval of $[x, x + t)$. We denote this by ${}_t d_x$.

That is,

$$ {}_t d_x = l_x - l_{x+t} $$

We can then calculate ${}_t q_x$ and ${}_{t|u} q_x$ by the following two relations:

$$ {}_t q_x = \frac{{}_t d_x}{l_x} = \frac{l_x - l_{x+t}}{l_x} = 1 - \frac{l_{x+t}}{l_x} $$

$$ {}_{t|u} q_x = \frac{{}_u d_{x+t}}{l_x} = \frac{l_{x+t} - l_{x+t+u}}{l_x} $$

When $t = 1$, we can omit the subscript t and write ${}_1 d$ as d_x

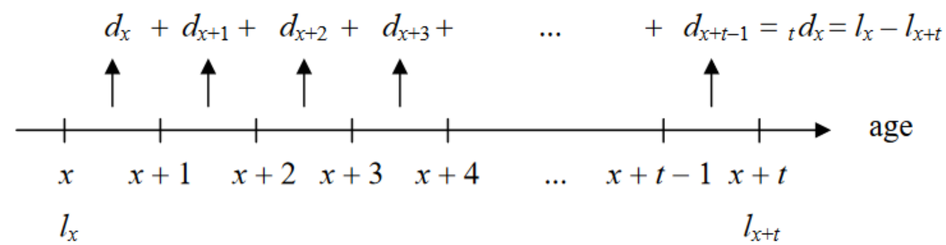
${}_1 d = d_x$ denotes the expected number of deaths between ages x and $x+1$

i.e. $d_x = l_x – l_{x+1}$

By definition, we have

$$ {}_t d_x = d_x + d_{x+1} + \ldots + d_{x+t-1} $$

Graphically



Also, when $t = 1$, we have the following relations:

$$ d_x = l_x – l_{x+1} $$

$$ p_x = \frac{l_{x+1}}{l_x} $$

$$ q_x = \frac{d_x}{l_x} $$

Summing up, with the life table functions l_x and d_x, we can recover survival probabilities ${}_t p_x$ and death probabilities ${}_t q_x$ for all integral values of t and x easily.

Life Table Functions

$${}_t p_x = \frac{l_{x+t}}{l_\mathrm{x}}$$

$${}_t d_x = l_x - l_{x+t} = d_x + d_{x+1} + \cdots + d_{x+t-1}$$

$${}_t q_x = \frac{{}_t d_x}{l_x} = \frac{l_x - l_{x+t}}{l_x} = 1 - \frac{l_{x+t}}{l_x}$$

$$p_x = \frac{l_{x+1}}{l_\mathrm{x}}$$

$$q_x = \frac{d_x}{l_x} = 1 - \frac{l_{x+1}}{l_x}$$

$${}_{t|u} q_x = \frac{l_{x+t} - l_{x+t+u}}{l_x}$$

Types of life tables

There are two types of life tables:

1) *Cohort* (or *generation*) life table and;

2) P*eriod* (or *current*) life table.

Cohort **(or *generation*) life table**

The cohort life table presents the mortality experience of a particular birth cohort – all persons born in the year 1900, for example – from the moment of birth through consecutive ages in successive calendar years.

It is usually not feasible to construct cohort life tables entirely on the basis of observed data for real cohorts due to data unavailability or incompleteness.

For example, a life table for a cohort of persons born in 1970 would require the use of data projection techniques to estimate deaths into the future.

Period **(or *current*) life table**

Unlike the cohort life table, the period life table does not represent the mortality experience of an actual birth cohort.

Rather, the period life table presents what would happen to a hypothetical cohort if it experienced throughout its entire life the mortality conditions of a particular period in time.

Example 1:

Given the following excerpt of a life table:

x	l_x	d_x
20	96178.01	99.0569
21	96078.95	102.0149
22	95976.93	105.2582
23	95871.68	108.8135
24	95762.86	112.7102
25	95650.15	116.9802

Calculate the following:

(a) ${}_5p_{20}$

(b) q_{24}

(c) ${}_{4|1}q_{20}$

Solution

(a) ${}_5p_{20} = \dfrac{l_{25}}{l_{20}} = \dfrac{95650.15}{96178.01} = 0.994512$.

(b) $q_{24} = \dfrac{d_{24}}{l_{24}} = \dfrac{112.7102}{95762.86} = 0.001177$.

(c) ${}_{4|1}q_{20} = \dfrac{{}_1d_{24}}{l_{20}} = \dfrac{112.7102}{96178.01} = 0.001172$.

Example 2:

Given

(i) $S_0(x) = 1 - \dfrac{x}{100}, \quad 0 \le x \le 100$

(ii) $l_0 = 100$

(a) Find an expression for l_x for $0 \le x \le 100$.

(b) Calculate q_2.

(c) Calculate ${}_3q_2$.

solution:

(a) $l_x = l_0\, S_0(x) = 100 - x$.

(b) $q_2 = \dfrac{l_2 - l_3}{l_2} = \dfrac{98-97}{98} = \dfrac{1}{98}$.

(c) ${}_3q_2 = \dfrac{l_2 - l_5}{l_2} = \dfrac{98-95}{98} = \dfrac{3}{98}$.

Example 3:

Consider the life table

Age	ℓ_x	d_x
0	100,000	501
1	99,499	504
2	98,995	506
3	98,489	509
4	97,980	512
5	97,468	514

(a) Find the number of individuals who die between ages 2 and 5.
(b) Find the probability of a life aged 2 to survive to age 4.

solution:

(a) ${}_2d_3 = \ell_2 - \ell_5 = 98,995 - 97,468 = 1527$

(b) ${}_2p_2 = \frac{\ell_4}{\ell_2} = \frac{97,980}{98,995} = 0.98975$

Example 4:

Given that: $l_x = 10000(100 - x)^2 \qquad 0 \leq x < 100$

Calculate the probability that a person now aged 20 will reach retirement age of 65.

Calculate the probability that a life (65) will die between ages 80 and 90

solution:

$${}_{45}p_{20} = \frac{\ell_{65}}{\ell_{20}} = \frac{10000(100-65)^2}{10000(100-20)^2} = 0.1914$$

$${}_{15|10}q_{65} = \frac{l_{80} - l_{90}}{l_{65}}$$

$${}_{15|10}q_{65} = \frac{[10000(100-80)^2 - 10000(100-90)^2}{10000(100-65)^2}$$

$${}_{15|10}q_{65} = \frac{[10000(20)^2 - 10000(10)^2}{10000(35)^2}$$

$${}_{15|10}q_{65} = \frac{[(20)^2-(10)^2}{(35)^2} = 0.2449$$

Optional Assignment 1

Complete the entries in the following table:

Age	l_x	d_x	p_x	q_x
0	100,000	·	·	·
1	97,523	·	·	·
2	94,123	·	·	·
3	91,174	·	·	·
4	87,234	·	·	·
5	85,938	–	–	–

Optional Assignment 2

Suppose that $S(x) = 1 - x/10 \qquad 0 \le x < 10$

(a) Find l_x:

(b) Using life table terminology, find p_2, q_3, ${}_3p_7$ and ${}_2q_7$:

Optional Assignment 2

The following is an extract from a life table.

Age	l_x
30	10,000
31	9965
32	9927
33	9885
34	9839
35	9789
36	9734
37	9673
38	9607
39	9534

Using the table above, find

(a) l_{36}

(b) d_{34}

(c) ${}_3d_{36}$

(d) ${}_5q_{30}$

(e) the probability that a life aged 30 dies between the ages 35 and 36.

Fractional Age Assumptions

Life tables only show values of l_x for x a non-negative integer, however, we may require mortality probabilities for non-integer ages

or

Given a life table, we can calculate values of ${}_t p_x$ and ${}_t q_x$ when both t and x are integers. But what if t and/or x are not integers?

In this case, we need to make an ***assumption about how the survival function behaves between two integral ages***.

We call such an assumption a ***fractional age assumption***.

There are two fractional age assumptions to be considered:

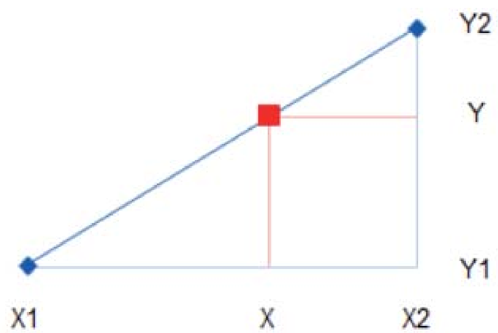
1. Uniform distribution of death
2. Constant force of mortality

Assumption 1: Uniform Distribution of Death (UDD)

The UDD assumption is the simplest of the fractional age assumptions and assumes that between integer-valued years, <u>the death rate is constant</u>.

Suppose we have $l_x > l_{x+t} > l_{x+1}$ with $x < x+t<x+1$, since we are assuming a constant death rate, linear interpolation can be used to obtain an expression for l_{x+t}.

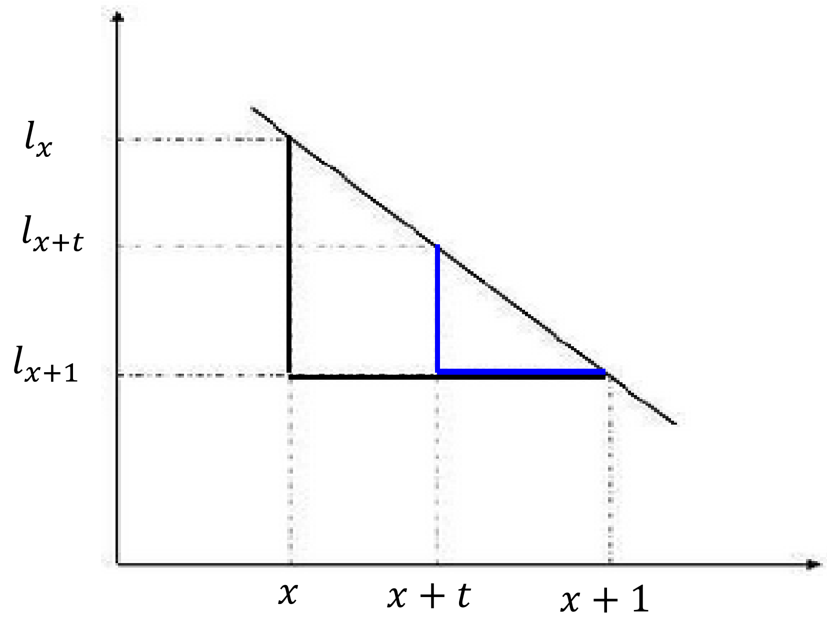
linear interpolation revisited



$$\frac{(X - X1)}{(X2 - X1)} = \frac{(Y - Y1)}{(Y2 - Y1)}$$

$$Y = Y1 + (X - X1)\frac{(Y2 - Y1)}{(X2 - X1)}$$

$$\text{Linear Interpolation}(y) = y_1 + (x - x_1)\frac{(y_2 - y_1)}{(x_2 - x_1)}$$



$$(l_{x+t} - l_{x+1})(x+1-x) = (l_x - l_{x+1})(x+1-(x+t))$$

$$(l_{x+t} - l_{x+1}) = (l_x - l_{x+1})(1-t)$$

$$l_{x+t} = l_x - t(l_x - l_{x+1})$$

$$l_{x+t} = (1-t)l_x - tl_{x+1}$$

$$\boxed{\Rightarrow\ l_{x+t} = l_x - td_x}$$

$$l_{x+t} = l_x - td_x$$

$$_tp_x = \frac{l_x - td_x}{l_x}$$

$$= 1 - tq_x$$

$$\boxed{_tp_x = 1 - tq_x}$$

$$1 - \ _tp_x = tq_x$$

$$\Rightarrow \quad _tq_x = tq_x$$

Key Equation for the UDD Assumption

$$\boxed{_tq_x = tq_x} \qquad 0 \leq t < 1$$

The equation

$$\boxed{l_{x+t} = (1-t)l_x - tl_{x+1}} \qquad 0 \leq t < 1$$

the interpolation between l_x and l_{x+1}

Is very useful if you are given a table of l_x (instead of q_x)

This equation $l_{x+t} = l_x - td_x$

is used to derive expressions for the *survival*, *distribution*, and *density functions*, and the *force of mo*rtality under the UDD assumption.

For integer x and $0 \leq t \leq 1$ we have

$$\boxed{_tp_x = \frac{l_{x+t}}{l_x} = \frac{l_x - td_x}{l_x} = 1 - \frac{td_x}{l_x} = 1 - tq_x}$$

$$\boxed{f_x(t) = -\frac{d \ _tp_x}{dt} = -\frac{d}{dt}(1 - tq_x) = q_x}$$

$$\boxed{\mu_{x+t} = \frac{f_t(x)}{_tp_x} = \frac{q_x}{1 - tq_x}}$$

$$\boxed{_tq_x = 1 - tp_x = tq_x}$$

- It is not necessary to learn all four of the expressions above since by knowing any one of them the others may be easily determined.
- The most useful expression is that for the distribution function as this is the simplest to apply when calculating probabilities.

<u>If the subscript on the left-hand-side is greater than 1</u>

First use equation $_{t+u}p_x = {}_tp_x \times {}_up_{x+t}$, to break down the probability into smaller portions.

<u>Example</u>: calculate $_{2.5}p_{30}$ as follows

$$_{2.5}p_{30} = {}_2p_{30} \times {}_{0.5}p_{32} = {}_2p_{30} \times (1 - 0.5q_{32}).$$

By applying $_tp_x = 1 - tq_x$

The value of $_2p_{30}$ and q_{32} can be obtained from the life table straight forwardly.

<u>If the subscript on the right-hand-side is not an integer</u>

Make use of the following:

Example: Let us consider $_{0.1}p_{5.7}$ (both subscripts are not integers).

First, multiply this probability with $_{0.7}p_5$, that is

$$_{0.7}p_5 \times {}_{0.1}p_{5.7} = {}_{0.8}p_5$$

This gives

$$_{0.1}p_{5.7} = \frac{_{0.8}p_5}{_{0.7}p_5} = \frac{1-0.8q_5}{1-0.7q_5}$$

The value of q_5 can be obtained from the life table.

Summary: Uniform Distribution of Death (UDD) $0 \le t < 1$

$$\boxed{l_{x+t} = l_x - td_x}$$

$$\boxed{_tq_x = 1 - tp_x = tq_x}$$

$$\boxed{l_{x+t} = (1-t)l_x - tl_{x+1}}$$

$$\boxed{_tp_x = \frac{l_{x+t}}{l_x} = \frac{l_x - td_x}{l_x} = 1 - \frac{td_x}{l_x} = 1 - tq_x}$$

$$\boxed{f_x(t) = -\frac{d\ {}_tp_x}{dt} = -\frac{d}{dt}(1 - tq_x) = q_x}$$

$$\boxed{\mu_{x+t} = \frac{f_t(x)}{_tp_x} = \frac{q_x}{1 - tq_x}}$$

Example 5

Given the following extract from a life table

x	l_x
35	97250
36	97126
37	96993

Under the UDD assumption, compute the probabilities

(i) ${}_{0.5}p_{35}$, (ii) ${}_{1.5}p_{35}$

Solution

(i) $${}_{0.5}p_{35} = 1 - {}_{0.5}q_{35} = 1 - 0.5q_{35}$$

where

$$q_{35} = 1 - p_{35} = 1 - \frac{l_{36}}{l_{35}} = 1 - \frac{97126}{97250} = 0.001275$$

Therefore

$${}_{0.5}p_{35} = 1 - (0.5 \cdot 0.001275) = 0.9993625$$

(ii) This equation can only be used when $0 \leq t \leq 1$ but in this case we have t =1.5.

We must instead first use the multiplicative property of the survival function to write the probability as

$${}_{t+u}p_x = {}_tp_x \times {}_up_{x+t}, \qquad \text{multiplicative property}$$

$${}_{1.5}p_{35} = {}_{0.5}p_{36} \cdot p_{35}$$

The UDD assumption can now be applied to evaluate ${}_{0.5}p_{36}$ while p_{35} can be calculated using standard results.

$${}_{0.5}p_{36} = 1 - {}_{0.5}q_{36} = 1 - 0.5q_{36} = 1 - 0.5\left(1 - \frac{l_{37}}{l_{36}}\right)$$

$$= 1 - 0.5\left(1 - \frac{96993}{97126}\right) = 0.9993$$

$$p_{35} = \frac{l_{36}}{l_{35}} = \frac{97126}{97250} = 0.9987$$

The overall probability is therefore

$${}_{1.5}p_{35} = 0.9993 \cdot 0.9987 = 0.9980$$

Assumption 2: Constant Force of Mortality

The assumption of a constant force of mortality is based on the idea of exponential interpolation and assumes that:

$$l_{x+t} = ab^t, \qquad 0 \le t \le 1.$$

We can determine the values of a and b by considering different values of t.

$$t = 0 \implies l_x = a$$

$$t = 1 \implies l_{x+1} = ab \implies b = \frac{l_{x+1}}{a} = \frac{l_{x+1}}{l_x}$$

Therefore

$$\boxed{l_{x+t} = l_x \left(\frac{l_{x+1}}{l_x}\right)^t = (l_x)^{1-t}(l_{x+1})^t}$$

This equation is one of the key equations for the constant force of mortality assumption and we now use it to derive expressions for the survival function, density function, and force of mortality.

$$\begin{aligned} {}_tp_x &= \frac{l_{x+t}}{l_x} = \frac{(l_x)^{1-t}(l_{x+1})^t}{l_x} \\ &= \left(\frac{l_{x+1}}{l_x}\right)^t \end{aligned}$$

$$\Rightarrow \quad {}_tp_x = (p_x)^t$$

$$\boxed{{}_tq_x = 1 - (p_x)^t}$$

To find the p.d.f we recall that when differentiating a^x with respect to x we instead apply the chain rule to *exp*(*xln* (*a*)), therefore,

$$\boxed{f_x(t) = -\frac{d\ {}_tp_x}{dt} = -\frac{d}{dt}(p_x)^t = -\frac{d}{dt}\exp(t\log(p_x)) - p_x^t \log(p_x)}$$

Finally, the force of mortality is given by

$$\boxed{\mu_{x+t} = \frac{f_x(x)}{{}_tp_x} = \frac{p_x^t \log(p_x)}{p_x^t} = -\log(p_x)}$$

Note that since $0 \le p_x \le 1$, $\log(p_x) \le 0$ and so the force of mortality and p.d.f are non-negative as required.

Summary: Constant Force of Mortality $0 \le t < 1$

$$\boxed{l_{x+t} = l_x \left(\frac{l_{x+1}}{l_x}\right)^t = (l_x)^{1-t}(l_{x+1})^t}$$

$$\boxed{{}_tq_x = 1 - (p_x)^t}$$

$$\boxed{{}_tp_x = (p_x)^t}$$

$$\boxed{f_x(t) = -\frac{d\ {}_tp_x}{dt} = -\frac{d}{dt}(p_x)^t = -\frac{d}{dt}\exp(t\log(p_x)) - p_x^t\log(p_x)}$$

$$\boxed{\mu_{x+t} = \frac{f_x(x)}{{}_tp_x} = \frac{p_x^t\log(p_x)}{p_x^t} = -\log(p_x)}$$

The central rate of mortality

We have seen that q_x represents the probability that a life of exact age *x* dies before reaching exact age (*x* + 1).

Then, q_x is often referred to as the initial rate of mortality at exact age.

An alternative definition of the rate of mortality is often used in demography.

We define the **central rate of mortality at exact age**, denoted by m_x, as follows:

$$m_x = \frac{\int_0^1 l_{x+t}\mu_{x+t}\,dt}{\int_0^1 l_{x+t}\,dt} = \frac{\int_0^1 {}_tp_x\mu_{x+t}\,dt}{\int_0^1 {}_tp_x\,dt} = \frac{q_x}{\int_0^1 {}_tp_x\,dt}$$

In practice, the central rate of mortality m_x represents a <u>weighted average of the force of mortality applying over the year of age *x* to (*x*+1)</u>, and can be thought as the <u>probability that a life alive between ages *x* and (*x*+1) dies before attaining exact age (*x*+1)</u>.

The importance of the central rate of mortality m_x arose because, historically, it was easier for actuaries to estimate this quantity from the observed data than either the initial rate of mortality, q_x, or the force of mortality, μ_x.

Example 6

Given the following extract from a life table.

x	l_x
80	53925
81	50987
82	47940
83	44803

Under the assumption of a constant force of mortality between integer ages, calculate the probabilities

(i) ${}_{0.75}p_{80}$, (ii) ${}_{2.25}p_{80}$

Solution

(i) Since $t \leq 1$, the equation can be applied straight away to give

$$\boxed{{}_tp_x = (p_x)^t} = \left(\frac{l_{x+1}}{l_x}\right)^t$$

$${}_{0.75}p_{80} = (p_{80})^{0.75} = \left(\frac{l_{81}}{l_{80}}\right)^{0.75} = \left(\frac{50987}{53925}\right)^{0.75} = 0.95885$$

(ii) we use the multiplicative property of the survival function to give

$${}_{t+u}p_x = {}_tp_x \times {}_up_{x+t}, \qquad \text{multiplicative property}$$

$${}_{2.25}p_{80} = {}_2p_{80} \cdot {}_{0.25}p_{82}$$

where

$${}_2p_{80} = \frac{l_{82}}{l_{80}} = \frac{47940}{53925} = 0.8890$$

and

$${}_{0.25}p_{82} = (p_{82})^{0.25} = \left(\frac{l_{83}}{l_{82}}\right)^{0.25} = \left(\frac{44803}{47940}\right)^{0.25} = 0.9832.$$

The total probability is therefore

$${}_{2.25}p_{80} = 0.8890 \cdot 0.9832 = 0.8741.$$

The following table, summarizes the formulas for the two fractional age assumptions

	UDD	Constant force
${}_r p_x$	$1 - r q_x$	$(p_x)^r$
${}_r q_x$	$r q_x$	$1 - (p_x)^r$
μ_{x+r}	$\dfrac{q_x}{1 - r q_x}$	$-\ln(p_x)$

In the table, x is an integer and $0 \le r < 1$.

The shaded formulas are the key formulas that you must remember

Example 7

Consider the life table

Age	ℓ_x	d_x
0	100,000	501
1	99,499	504
2	98,995	506
3	98,489	509
4	97,980	512
5	97,468	514

(a) Find the number of individuals who die between ages 2 and 5.
(b) Find the probability of a life aged 2 to survive to age 4.

Solution

(a) ${}_2d_3 = \ell_2 - \ell_5 = 98,995 - 97,468 = 1527$

(b) ${}_2p_2 = \frac{\ell_4}{\ell_2} = \frac{97{,}980}{98{,}995} = 0.98975$

Curtate expectation of life

Similarly, the curtate expectation of life is given by

$$e_x = \sum_{k=1}^{\infty} {}_k p_x = \sum_{k=1}^{\infty} \frac{l_{x+k}}{l_x}$$

In addition the temporary curtate expectation of life is defined by

$$e_{x:\overline{n}|} = \sum_{k=1}^{n} \frac{l_{x+k}}{l_x}$$

x	80	81	82	83	84	85	86
l_x	250	217	161	107	62	28	0

(1) Calculate d_x, x = 80, 81…; 86.

(2) Calculate the probability mass function of the curtate life K_{80}.

(3) Calculate e_{80} and $Var(K_{80})$ using the probability mass function of K_{80}.

(4) Calculate e_{80} and $Var(K_{80})$ using.

(5) Calculate $e_{80:\overline{3}|}$

Mean and Variance of X

The mean of an age-at-death random variable is given by

$$\mathring{e}_0 = E(X) = \int_0^{\infty} s(x)dx = \int_0^{\infty} {}_x p_0 dx.$$

we can express the mean in terms of life table terms

$$\mathring{e}_0 = \frac{1}{\ell_0} \int_0^{\infty} \ell_x dx.$$

Select-and-Ultimate Tables

The life tables we have employed thus far are strictly based on attained age. To incorporate the effects of underwriting on mortality, we can consider select and ultimate models.

Before being accepted for life assurance cover, potential policyholders are often required to undergo a medical examination to satisfy the insurer that they are in a 'reasonable' level of health.

Lives who fail to satisfy the requirements laid down by the insurance company will often be refused cover (or required to pay a higher premium for the same level of cover).

For this reason, a person who has just purchased a life insurance cover can be expected to be in better health and thus has a lower probability of death than a person the same age in the general population.

This effect is known as **selection** (i.e. the process of choosing lives for membership of a defined group, rather than random sampling)

However, the effect of underwriting will not last forever. The period after which the effect of underwriting is completely gone is called the **select period**.

Once the duration since selection exceeds the select period, the lives are assumed to experience the **ultimate mortality rates** appropriate for the general population at the same age.

Construction and application of a select life table

A mortality table for the insured population must therefore consider both the age of issue and the duration since issue.

We make the following modeling assumptions:

(1) Survival depends on the following parameters.

(a) the age x at which this individual joined the insured group -
(b) the current age of the person

(2) Suppose the select period is d years after the issue.

The parameters are then written as subscripts with a square bracket around the first parameter and a plus sign between the parameters.

In other words, the subscript is of the form $[x]+t$. When t = 0, it is omitted.

$[x]$ indicates the age at selection (i.e., the age at which the underwriting was done).

$[x]+t$ indicates a person currently age $x+t$ and was selected at age x (i.e., the underwriting was done at age x). This implies that the insurance contract has elapsed for t years.

Thus the mortality rate for a 60-year-old who just purchased a life insurance policy would be written as $q_{[60]}$

The mortality rate for a 65-year-old who purchased a policy at age 60 would be written $q_{[60]+5}$

When mortality depends on the initial age as well as duration, it is known as *select* mortality, since the person is *selected* at that age.

For example, we have the following select probabilities:

$q_{[x]}$ is the probability that a life age x now dies before age $x+1$, given that the life is selected at age x.

$q_{[x]+t}$ is the probability that a life age $x+t$ now dies before age x+t+1, given that the life was selected at age x.

Suppose that the select period is d years, we have

$q_{[x]+t} < q_{x+t}$, for $t < d$, for t = 0,1,2,..., d -1
$q_{[x]+t} = q_{x+t}$, for $t \geq d$.

When the life is no longer in the select period we drop the square brackets and revert back to the original notation.

The ordinary death probability q_{x+t} is called the ultimate death probability.

A life table that contains both select probabilities and ultimate probabilities is called a select-and-ultimate life table.

An individual aged x, included in the experience, can be represented by

[x]	if just entering select status,
[x-1]+1	if select status was entered 1 year ago, and generally,
[x - t]+*t*	if select status was entered *t* years ago.

This notation is used in conjunction with the standard life table functions.

For example,

q[x] is the probability that an individual aged *x*, who has just entered select status, will not survive till age *(x* + 1).

Similarly

q[x]+*t* is the probability that an individual who entered select status at age *x*, and has survived *t* years, will not survive one further year.

So that

q_x is the ultimate mortality rate (the probability that a life x, chosen from the general population, will die before age x+1); and

$q_{[x]}$ is the select mortality rate (the probability that a life x, chosen from the selected insurance population, will die before age x+1)

To represent a select and ultimate life table, we use a two-dimensional array. Each different row represents a different age at selection, and each column represents the duration since the selection

In general:

The select table corresponds to the first d columns, and the ultimate table corresponds to the d+1 column.

Suppose that a select period is three years. Then, a select and ultimate life table has the form

	Select period=3			ultimate	
$[x]$	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	ℓ_{x+3}	$x+3$
$[1]$	$\ell_{[1]}$	$\ell_{[1]+1}$	$\ell_{[1]+2}$	ℓ_4	4
$[2]$	$\ell_{[2]}$	$\ell_{[2]+1}$	$\ell_{[2]+2}$	ℓ_5	5
$[3]$	$\ell_{[3]}$	$\ell_{[3]+1}$	$\ell_{[3]+2}$	ℓ_6	6

Let $q_{[x-d]+d}$ represents the mortality rate of a person aged x who joined the select group d years ago, then

$$q_{[x]} < q_{[x-1]+1} < q_{[x-2]+2} < \ldots < q_{[x-(d-1)]+(d-1)} < q_{[x-d]+d}.$$

If d is the select period, then

$$q_{[x-d]+d} = q_x.$$

Therefore, in order to find the mortality rate:

Step1: Find the age at selection $[x]$.

Step 2: In the corresponding row, move right until you cross the desired probability.

Step 3: If the duration is longer than the selection period d, you will not get this probability on the chosen row, so you have to move down at the end of the row.

Computing mortality for issue age 20 is shown schematically in the following figure:

x	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$	q_{x+3}	$x+3$
18					21
19					22
20					23
21					24
22					25
23					26
24					27

- Notice that there is no direct way to go from [x] to [x + 1].
- A person who is selected at age x will be [x] in the first year, [x]+1 in the second year, and so on, until he becomes ultimate at age $x+d$, and then his age will continue to increase and be without brackets.
- He/She will never be selected again. He/She will never be $[x+l]$.

A life selected at age x can never become a life selected at any higher age. $[x]$ will never become $[x+1]$.

If you need to relate $[x]$ and $[x+1]$, the only way to do it is to go through the ultimate table.

Sometimes, you may be given a select-and-ultimate table that contains the life table function ℓ_x.

In this case, you can calculate survival and death probabilities by using the following equations:

Suppose that we start with $\ell_{[x]}$ lives that entered the select period at age x: Then the number of survivors at time t is denoted by $\ell_{[x]+t}$.

We denote by

$$_np_{[x]+t} = \frac{\ell_{[x]+t+n}}{\ell_{[x]+t}} \text{ and } _nq_{[x]+t} = \frac{\ell_{[x]+t} - \ell_{[x]+t+n}}{\ell_{[x]+t}}.$$

When n = 1, we use the notation

$$p_{[x]+t} = \frac{\ell_{[x]+t+1}}{\ell_{[x]+t}} \text{ and } q_{[x]+t} = \frac{\ell_{[x]+t} - \ell_{[x]+t-1}}{\ell_{[x]+t}}$$

Notice that

$$p_{[x]}p_{[x]+1}\cdots p_{[x]+n-1} = \frac{\ell_{[x]+1}}{\ell_{[x]}}\frac{\ell_{[x]+2}}{\ell_{[x]+1}}\cdots,\frac{\ell_{[x]+n}}{\ell_{[x]+n-1}} = \frac{\ell_{[x]+n}}{\ell_{[x]}} = {}_np_{[x]}$$

Example 8

Consider the following select table:

x	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$
35	0.013	0.012	0.011

If $\ell_{[35]} = 1000$, *find* $\ell_{[35]+1}$, $\ell_{[35]+2}$ *and* $\ell_{[35]+3}$

Solution 8

From $p_{[x]+t} = \frac{\ell_{[x]+t+1}}{\ell_{[x]+t}}$, we get $\ell_{[x]+t+1} = \ell_{[x]+t} p_{[x]+t}$.

Hence

$$\ell_{[35]+1} = \ell_{[35]} p_{[35]} = (1000)(1 - 0.013) = 987,$$
$$\ell_{[35]+2} = \ell_{[35]+1} p_{[35]+1} = (987)(1 - 0.012) = 975.156,$$
$$\ell_{[35]+3} = \ell_{[35]+2} p_{[35]+2} = (975.156)(1 - 0.011) = 964.429284$$

We get

x	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	$\ell_{[x]+3}$
35	1000	987	975.156	964.429284

Example 9

You are given the following extract from a 2–year select–and–ultimate mortality table:

$[x]$	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
45	1235	1124	1039	47
46	1135	1025	978	48
47	1012	996	965	49

(i) Complete the table

$[x]$	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
45				47
46				48
47			–	49

(ii) Find ${}_2p_{[47]}$, ${}_2p_{[46]+1}$ and ${}_2p_{47}$.

Solution 9

(i)

$[x]$	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
45	$\frac{1235-1124}{1235}$	$\frac{1124-1039}{1124}$	$\frac{1039-978}{1039}$	47
46	$\frac{1135-1025}{1135}$	$\frac{1025-978}{1025}$	$\frac{978-965}{978}$	48
47	$\frac{1012-996}{1012}$	$\frac{996-965}{996}$	—	49

(ii)

$$_2p_{[47]} = \frac{\ell_{49}}{\ell_{[47]}} = \frac{965}{1012} = 0.9535573123$$

$$_2p_{[46]+1} = \frac{\ell_{49}}{\ell_{[46]+1}} = \frac{965}{1025} = 0.9414634146$$

$$_2p_{47} = \frac{\ell_{49}}{\ell_{47}} = \frac{965}{1039} = 0.9287776708$$

Example 10

You are given the following extract from a 2–year select–and–ultimate mortality table:

$[x]$	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
45	0.009	0.008	0.007	47
46	0.008	0.006	0.005	48

Complete the table

$[x]$	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
45	10000			47
46				48

Solution 10

$$\ell_{[45]+1} = \ell_{[45]}p_{[45]} = (10000)(1-0.009) = 9910,$$

$$\ell_{47} = \ell_{[45]+1}p_{[45]+1} = (10000)(1-0.009)(1-0.008) = 9830.72,$$

$$\ell_{48} = \ell_{47}p_{47} = (10000)(1-0.009)(1-0.008)(1-0.007) = 9761.90496,$$

$$\ell_{[46]+1} = \frac{\ell_{[48]}}{p_{[46]+1}} = \frac{(10000)(1-0.009)(1-0.008)(1-0.007)}{(1-0.006)} = 9820.82$$

$$\ell_{[46]} = \frac{\ell_{[46]+1}}{p_{[46]}} = \frac{(10000)(1-0.009)(1-0.008)(1-0.007)}{((1-0.006)(1-0.008)} = 9900.0301$$

$[x]$	$\ell_{[x]}$	$\ell_{[x]+1}$	ℓ_{x+2}	$x+2$
45	10000	9910	9830.72	47
46	9900.030181	9820.82994	9761.90496	48

Example 11:

The following is an excerpt of a (hypothetical) select-and-ultimate table with a select period of two years.

x	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
40	0.04	0.06	0.08	42
41	0.05	0.07	0.09	43
42	0.06	0.08	0.10	44
43	0.07	0.09	0.11	45

Let us consider a person who is currently age 41 and is just selected. The death probabilities for this person are as follows:

Age 41: $q_{[41]} = 0.05$

Age 42: $q_{[41]+1} = 0.07$

Age 43: $q_{[41]+2} = q_{43} = 0.09$

Age 44: $q_{[41]+3} = q_{44} = 0.10$

Age 45: $q_{[41]+4} = q_{45} = 0.11$

let us consider a person who is currently age 41 and was selected at age 40. The death probabilities for this person are as follows

Age 41: $q_{[40]+1} = 0.06$

Age 42: $q_{[40]+2} = q_{42} = 0.08$

Age 43: $q_{[40]+3} = q_{43} = 0.09$

Age 44: $q_{[40]+4} = q_{44} = 0.10$

Age 45: $q_{[40]+5} = q_{45} = 0.11$

Even though the two persons considered are of the same age now, their current death probabilities are different.

Because the first individual has the underwriting done more recently, the effect of underwriting on him/her is stronger, which means he/she should have a lower death probability than the second individual.

Example 12:

Given the following select-and-ultimate table with a 2-year select period:

x	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
30	9907	9905	9901	32
31	9903	9900	9897	33
32	9899	9896	9892	34
33	9894	9891	9887	35

Calculate the following:

(a) ${}_2q_{[31]}$

(b) ${}_2p_{[30]+1}$

(c) ${}_{1|2}q_{[31]+1}$

Solution 12

(a) $${}_2q_{[31]} = 1 - \frac{l_{[31]+2}}{l_{[31]}} = 1 - \frac{l_{33}}{l_{[31]}} = 1 - \frac{9897}{9903} = 0.000606$$

(b) $${}_2p_{[30]+1} = \frac{l_{[30]+1+2}}{l_{[30]+1}} = \frac{l_{33}}{l_{[30]+1}} = \frac{9897}{9905} = 0.999192.$$

(c) $${}_{1|2}q_{[31]+1} = \frac{l_{[31]+1+1} - l_{[31]+1+1+2}}{l_{[31]+1}} = \frac{l_{33} - l_{35}}{l_{[31]+1}} = \frac{9897 - 9887}{9900} = 0.001010$$

Example 13:

Consider the following extract from a select life table

x	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$
35	0.013	0.012	0.011

If $l_{[35]} = 1000$, find $l_{[35]+1}$ and $l_{[35]+2}$.

Solution 13

We know $$p_{[35]} = \frac{l_{[35]+1}}{l_{[35]}}$$

$$\Longrightarrow \quad l_{[35]+1} = p_{[35]} \cdot l_{[35]} = (1 - q_{[35]}) \cdot l_{[35]}$$

$$= (1 - 0.013) \cdot 1000 = 987$$

and hence

$$p_{[35]+1} = \frac{l_{[35]+2}}{l_{[35]+1}}$$

$$\Longrightarrow \quad l_{[35]+2} = p_{[35]+1} \cdot l_{[35]+1} = (1 - q_{[35]+1}) \cdot l_{[35]+1}$$

$$= (1 - 0.012) \cdot 987 = 975.16$$

It is also possible to set questions to examine your knowledge on *select-and-ultimate tables and fractional age assumptions at the same time*.

An example which involves a select-and-ultimate table and the UDD assumption.

Example 14:

Consider the following two-year select life table

x	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
45	1235	1124	1039	47
46	1135	1025	978	48
47	1012	996	965	49

Calculate

(i) $_{2}p_{[46]+1}$ and

(ii) $_{0.4}p_{[45]+1.7}$, using the UDD assumption for the second part.

Solution 14

(i) Notice that since the select period is 2 years, after 3 years the life will no longer be select and $l_{[x]+3} = l_{x+3}$, then

$$_{2}p_{[46]+1} = \frac{l_{[46]+3}}{l_{[46]+1}} = \frac{l_{49}}{l_{[46]+1}} = \frac{965}{1025} = 0.9415$$

(ii)

$$_{0.4}p_{[45]+1.7} = \frac{l_{[45]+2.1}}{l_{[45]+1.7}} = \frac{l_{47.1}}{l_{([45]+1)+0.7}}$$

Now, using $\boxed{l_{x+t} = l_x - td_x}$

and considering the numerator and denominator separately

$$\begin{aligned} l_{47.1} &= l_{47} - 0.1(l_{47} - l_{48}) \\ &= 1039 - 0.1(1039 - 978) = 1032.9 \end{aligned}$$

and

$$\begin{aligned} l_{([45]+1)+0.7} &= l_{[45]+1} - 0.7(l_{[45]+1} - l_{[45]+2}) \\ &= l_{[45]+1} - 0.7(l_{[45]+1} - l_{47}) \\ &= 1124 - 0.7(1124 - 1039) \\ &= 1064.5 \end{aligned}$$

Therefore, the overall probability is

$$_{0.4}p_{[45]+1.7} = \frac{1032.9}{1064.5} = 0.9703$$

Optional Assignment 1

Given the following select-and-ultimate table with a select period of 2 years:

x	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
70	22507	22200	21722	72
71	21500	21188	20696	73
72	20443	20126	19624	74
73	19339	19019	18508	75
74	18192	17871	17355	76

(a) Compute ${}_3p_{73}$.

(b) Compute the probability that a life age 71 dies between ages 75 and 76, given that the life was selected at age 70.

(c) Assuming uniform distribution of deaths between integral ages, calculate ${}_{0.5}p_{[70]+0.7}$.

(d) Assuming constant force of mortality between integral ages, calculate ${}_{0.5}p_{[70]+0.7}$.

Optional Assignment 2

You are given the following extract from a 2-year select and-ultimate mortality table:

$[x]$	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
45	1235	1124	1039	47
46	1135	1025	978	48
47	1012	996	965	49

(i) Complete the table

$[x]$	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
45				47
46				48
47			-	49

(ii) Find ${}_2p_{[47]}$, ${}_2p_{[46]+1}$ and ${}_2p_{47}$.

END